STAFF REVIEW · H446 020 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. The third element, because array indexing starts at zero.

2. An array has a fixed length after creation; a List<T> can grow and shrink.

3. The location or index of the first node in the logical sequence.

4. 40 → 10 → 30 → 20, then -1 ends the traversal.

I Do — source worked steps

1. Store 8 — CODE:
top = 0
stack[0] = 8

2. Store 7 — CODE:
top = 1
stack[1] = 7

3. Remove two values — CODE:
first = 7
second = 8
top = -1

4. Store result — CODE:
top = 0
stack[0] = 15

We Do — guided CHECK

1. Complete the same state
CODE:
Start: empty
push 8 → [8]
push 7 → [8, 7]
A: pop two, add, push → [15]

2. Complete the comparison
CODE:
Stack = ?
Queue = ?
CODE:
Stack = LIFO
Queue = FIFO

3. Fill the same code
CODE:
top = ?;
stack[top] = value;
CODE:
top = top + 1;
stack[top] = value;

4. Fill the same code
CODE:
value = queue[?];
front = ?;
CODE:
value = queue[front];
front = front + 1;

You Do — source mark schemes

1. CODE:
int[] stack = new int[5];
int top = -1;
top++; stack[top] = 5;
top++; stack[top] = 8;
top++; stack[top] = 3;
Console.WriteLine(stack[top]); // 3
top--;
Console.WriteLine(stack[top]); // 8
The top pointer becomes 1, pointing to 8.

2. CODE:
int[] queue = new int[5];
int front = 0;
int rear = -1;
rear++; queue[rear] = 10;
rear++; queue[rear] = 20;
rear++; queue[rear] = 30;
Console.WriteLine(queue[front]); // 10
front++;
The front pointer becomes 1, pointing to 20.

3. CODE:
int[] stack = { 2, 4, 6 };
int top = 2;
Console.WriteLine(stack[top]); // 6

top--;
int[] queue = { 2, 4, 6 };
int front = 0;
Console.WriteLine(queue[front]); // 2
front++;

Stack: last in, first out (6). Queue: first in, first out (2).

4. Stack underflow occurs when a pop operation is attempted on an empty stack (top = -1). The program should check whether the stack is empty before reading an array element. Otherwise it may access an invalid index or produce an error.

Exit Ticket — source answer

1. True. A stack is last-in first-out; a queue is first-in first-out.